Asthra Meadipudi

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OBJECTIVE

First-year SJSU student (Class of 2030) who builds and deploys data and machine learning projects with a focus on responsible AI. Brings 4+ years of teaching and mentoring experience and is eager to lead teams building ethical technology.

EDUCATION

San José State University, San Jose, CA **Expected May 2030**
Coursework: Python Programming (CS 22A), Technology Ethics (ENGR 25)

Dublin High School, Engineering and Design Academy, Dublin, CA Jun 2026
Coursework: AP Computer Science A, AP CS Principles, AP Statistics, AP Calculus AB

SKILLS & CERTIFICATIONS

Languages: Python, SQL, R, Java, JavaScript, C#
Data: pandas, scikit-learn, Keras (TensorFlow), Fairlearn, SHAP, Matplotlib, DuckDB
Tools: Git, GitHub, Jupyter, Streamlit, VS Code, command line (bash)
Certification: IBM Data Science Professional Certificate, IBM / Coursera (Sep 2026)

PROJECTS

Student Dropout Risk & Responsible AI Audit | GitHub | Live App Sep 2026 - Oct 2026

- Predicted student dropout risk on 4,424 real records (UCI dataset) by comparing logistic regression, random forest, neural network models in **Python**, **SQL**, **scikit-learn**, **Keras**, reaching AUC 0.91.
- Audited model fairness across 6 student groups using **Fairlearn**, **SHAP**, cutting age-based false alarm gap 38% with no recall loss.
- Deployed live risk-ranking web app with **Streamlit**, surfacing top factors behind each student's prediction.

E-Commerce Retention Analysis | GitHub Oct 2026

- Cleaned raw e-commerce event data into analysis-ready tables with **DuckDB**, defining core product metrics.
- Analyzed customer behavior with **SQL** queries, measuring retention, funnel conversion, repeat purchases.

EXPERIENCE

Teaching Assistant Intern, Data Science Lab with R | UC Berkeley ATDP | Berkeley, CA Jun 2025 - Aug 2025
UC Honors-designated, college-level data science and machine learning course for advanced high schoolers

- Mentored 45 students through end-to-end machine learning projects in **R**, guiding data cleaning, EDA, train/test splits, model tuning through final presentations.
- Taught linear regression plus classification models (logistic regression, LDA, QDA, Naive Bayes, KNN) using real-world examples, clarifying each model's assumptions.
- Explained hypothesis testing, p-values, confidence intervals, Central Limit Theorem in one-on-one sessions, tailoring instruction to each student's background.
- Streamlined grading by debugging student **R** code, setting up **GitHub** repositories, giving instructors access to all final projects.
- Led discussions on privacy, bias, p-hacking, responsible AI use, building ethical awareness in data analysis.

Teaching Assistant Intern, Digital Inventions | UC Berkeley ATDP | Berkeley, CA Jun 2025 - Jul 2025

- Coached 30 4th-5th graders one-on-one on variables, loops, conditionals in **Scratch**, helping each build original games, animations.

Code Sensei (Coding Instructor & Mentor) | Code Ninjas | Pleasanton, CA Dec 2021 - May 2026

- Taught coding to ~80 students ages 5 to 14 across CREATE, JR programs, logging 800+ instruction hours over 4.5 years.
- Advanced CREATE students (ages 8 to 14) from block coding to text languages using **MakeCode**, **JavaScript**, **C#** in **Unity**, **GDScript** in **Godot**, enabling them to build their own games.
- Introduced coding to JR students (ages 5 to 7) through hands-on visual activities in **ScratchJr**, **CodeSpark**, building early programming foundations.
- Taught students to debug independently by reading error messages, adapting instruction to each student's pace, learning style.
- Graded student game projects with specific code, design feedback; led summer camps, coding birthday parties.

ACTIVITIES

- **Responsible Computing Club, SJSU:** LegalBee team member; Mentorship Program mentee; attended 4 events
- **Girls Who Code and Society of Women Engineers, SJSU** (Fall 2026 - Present)